

Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

Abstract

CCS Concepts

- Human-centered computing → Mixed / augmented reality;
- Computing methodologies → Rendering; Computer vision.

Keywords

eye tracking, deep learning, computer graphics

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1 Introduction

2 Related works

To compare our work with existing eye convergence and accommodation tracking, we review related works from two main areas: physical feature-based experimental designs and virtual eye modeling and neural rendering. To clearly show the technical features of previous works, Table 1 categorizes and compares these methods based on their hardware setups, algorithms, operating modes, and tracking accuracy.

2.1 Eye Accommodation Tracking

In stereoscopic and near-eye displays, the **Vergence-Accommodation Conflict (VAC)** [Hoffman et al. 2008] arises when the eyes converge to a virtual 3D depth while the crystalline lens remains focused on the physical display plane. This mismatch can reduce fusion performance and cause visual discomfort. Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. However, vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems. Higher-order Purkinje images provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al. [2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al. [2023] extend this direction by collecting multi-subject

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Purkinje measurements and combining a Purkinje imaging setup with a **Multi-Layer Perceptron (MLP)**-based regression model to estimate dynamic vergence and accommodation across users.

These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye geometry. Motivated by this limitation, we first build a controllable 3D optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

2.2 Computational Eye Modeling

Computational eye models are widely used to generate labeled data for gaze estimation. UnityEyes, NVGaze, and RIT-Eyes synthesize controllable variations in gaze, ocular geometry, appearance, and illumination for training eye-tracking models [Kim et al. 2019; Nair et al. 2020; Wood et al. 2016]. These systems capture visible eye-region cues effectively, but they primarily target eye rotation, pupil localization, and segmentation rather than accommodation-dependent changes within the crystalline lens.

Neural eye representations improve visual realism and relighting. EyeNeRF combines an explicit eyeball surface with implicit volumetric representations for view synthesis, gaze animation, corneal refraction, and relighting [Li et al. 2022]. Khademi et al. extend a similar hybrid formulation to near-field point sources and structured illumination [Khademi et al. 2026]. However, these methods focus on photorealistic appearance and do not explicitly predict P3/P4 formation or accommodation-dependent internal ocular geometry.

Purkinje-based systems model internal reflections more directly. Wu et al., Idrizović et al., and OpenIrisDPI use P1 and P4 for accurate eye-rotation tracking [Idrizović et al. 2025; Ressmeyer et al. 2026; Wu et al. 2023]. Lu et al. further model higher-order reflections with multiple cameras and infrared LEDs to estimate vergence and accommodation [Lu et al. 2020]. Ozhan et al. simulate P1–P4 across accommodation and vergence states, validate the predictions against real captures, and use the resulting features for dynamic estimation [Ozhan et al. 2023]. Our work follows this direction by using a controllable optical eye model to predict how accommodation, vergence, and LED–camera geometry affect P1–P4 formation.

2.3 Physics-Based Optical Design

Differentiable optical design provides a possible framework for optimizing the illumination and imaging layout of a Purkinje capture system. Na et al. show that image-domain losses can be propagated through differentiable ray tracing to physical optical parameters [Na et al. 2024]. For our system, the same principle could be applied to LED position, camera pose, focal distance, aperture, and field of view.

The optimization criteria should follow the capture behavior of higher-order Purkinje reflections. Lu et al. show that P3 is weaker than P1, lies on a different focal plane, and may leave the camera

Table 1: Comparison of different eye and accommodation tracking methodologies. Accuracy is categorized into High (High), Mid (Mid, representing accuracy between 0.5° and 1.5°), and Low (Low).

No.	Ref.	Camera	Illum.	Method	Algorithm	Target	Mode	Accuracy
1	[Cornsweet and Crane 1973]	Multiple	IR	DPI	Geometric	Vergence	Desktop	Mid
2	[Morimoto et al. 2002]	Single	IR	PCCR	Geometric	Vergence	Desktop	Mid
3	[Sturgeon et al. 2025]	Multiple	IR	PCCR	Geometric	Vergence	Near-eye	Mid
4	[Bontha and Arefin 2024]	Multiple	IR	Mocap	Geometric	Vergence	Near-eye	Low
5	[Lu et al. 2020]	Multiple	IR	Purkinje	Learning	Both	Near-eye	High
6	[Ozhan et al. 2023]	Single	IR	Purkinje	Hybrid	Both	Desktop	High
7	[Nair et al. 2020]	Single	None	Ray Tracing	Rendering	Vergence	Synthetic	Mid
8	[Li et al. 2022]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low
9	[Khademi et al. 2026]	Multiple	Visible	NeRF	Learning	Vergence	Near-eye	Low

field of view as gaze changes; their system therefore uses multiple cameras and infrared LEDs [Lu et al. 2020]. Ozhan et al. further show that the camera-LED-eye geometry affects the observed P3/P4 locations and configure the system to improve reflection separation while avoiding vignetting [Ozhan et al. 2023]. These results suggest optimizing P3/P4 visibility, sharpness, field-of-view coverage, and separation from other reflections.

The selected layout must also remain practical to assemble, calibrate, and evaluate at the sensor level. Prior work on replicating a controllable AR haploscope highlights the implementation challenges of physical optical platforms [Bontha and Arefin 2024] and Wu et al. show that reliable P1/P4 measurement depends on sufficient image resolution and accurate reflection localization [Wu et al. 2023]. These considerations motivate evaluating optical layout, calibration, and sensor-level detectability together.

The present work does not perform end-to-end differentiable optimization. Instead, it uses geometric-optics simulation to compare candidate LED-camera configurations and to predict whether P1-P4 remain visible, sharp, and separable under realizable capture conditions. Differentiable layout optimization and synthetic-to-real adaptation are left as future extensions.

3 Optical Background

3.1 Accommodation-Dependent Purkinje Image Formation

The four principal Purkinje images are produced by specular reflections at the major refractive interfaces of the eye. P1 is formed at the anterior corneal surface, P2 at the posterior corneal surface, P3 at the anterior crystalline-lens surface, and P4 at the posterior crystalline-lens surface. The observed position of a higher-order reflection is not determined by its reflecting surface alone. Before reaching the camera, the corresponding rays pass through and are refracted by preceding ocular interfaces; their image-space positions therefore depend on the complete optical path through the eye.

Accommodation changes the optical power of the eye by altering crystalline-lens geometry. In accommodation-dependent schematic eye models, these changes are represented through lens-surface curvature, lens thickness, lens position, asphericity, and effective refractive index [Navarro et al. 1985]. P3 and P4 therefore respond

more directly to accommodation than the corneal reflections P1 and P2. Their displacement and magnification also depend on eye rotation and on the relative positions of the illumination source, eye, and camera [Lu et al. 2020; Ozhan et al. 2023].

The existence of a valid reflection path does not guarantee a usable image feature. P3 and P4 generally have lower contrast than P1 and are more sensitive to defocus, partial occlusion, competing glints, and sensor noise. Camera aperture and focal distance determine their spot size and sharpness at the sensor, while the pupil, iris, and eyelids restrict which reflection paths remain visible. Their detectability is therefore a joint property of ocular geometry, illumination, camera placement, and sensor configuration.

3.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1-P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [???]. Existing Purkinje imaging systems also provide practical guidance for hardware geometry. Both Lu et al. and Ozhan et al. place infrared LEDs close to the camera optical axis to maximize Purkinje visibility while keeping the reflections inside the camera field of view. At the same time, camera positions must balance the visibility of higher-order Purkinje images against eyelid and eyelash occlusions, glint overlap, and sensor saturation. These observations suggest that LED and camera placement is not arbitrary but jointly determines the brightness, separation, and detectability of P1-P4. Once the capture geometry is determined, synthetic rendering provides an efficient way to evaluate different configurations before physical experiments. RIT-Eyes demonstrates that physically based rendering can generate realistic near-eye images under controllable illumination, while EyeNeRF and subsequent hybrid eye representations further show that explicit corneal geometry is necessary for reproducing specular

reflections and refraction, whereas implicit neural representations are better suited for modeling complex periocular appearance and relighting. These observations motivate the use of explicit optical eye models for Purkinje simulation before subsequent NeRF or 3D Gaussian reconstruction, fall outside the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023]

4 Results and Discussions

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